

MACKAR INLET, NU, CanadaWMO: **710800**Lat: **68.300N**Lon: **85.670W**Elev: **1295**StdP: **14.02**Time Zone: **-6.00 (NAC)**Period: **82-92**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.8	-42.1	-41.8	0.5	-32.0	-39.8	0.6	-32.2	N/A	N/A	N/A	N/A	N/A	N/A	1.123

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	60.4	50.8	56.8	48.5	53.1	46.7	51.6	58.8	49.1	55.7	46.7	52.9	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
47.0	50.0	54.2	44.4	45.2	52.0	42.1	41.4	48.9	21.6	58.8	20.2	55.8	18.9	53.1	58.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-49.0	64.2	3.5	3.6	-51.5	66.8	-53.5	68.8	-55.5	70.8	-58.0	73.4
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-26.9	-26.2	-19.5	-4.6	13.4	32.6	44.0	36.8	27.5	10.7	-7.7	-19.4
	DBStd	26.65	10.41	10.83	10.81	10.81	9.22	7.54	8.09	6.75	7.65	10.32	11.67	13.00
	HDD50	16386	2384	2134	2153	1639	1135	523	226	416	676	1217	1731	2152
	HDD65	21813	2849	2554	2618	2089	1600	972	652	873	1126	1682	2181	2617
	CDD50	47	0	0	0	0	0	1	38	8	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	7.4	0.1	0.1	0.1	0.3	0.7	0.6	1.3	1.7	1.2	0.9	0.4	0.1
(15) Precipitation	PrecMax	11.7	0.7	0.5	1.6	0.9	2.0	1.6	3.5	3.0	2.5	2.0	1.3	0.6
	PrecMin	3.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.3	0.1	0.0	0.0
	PrecStd	1.7	0.2	0.1	0.3	0.2	0.5	0.4	0.8	0.8	0.5	0.6	0.4	0.2
	PrecStd	1.7	0.2	0.1	0.3	0.2	0.5	0.4	0.8	0.8	0.5	0.6	0.4	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	7.3	12.5	24.4	35.6	54.8	65.1	64.5	51.5	37.2	24.8	18.0
		MCWB	17.6	6.4	11.9	23.8	34.1	48.1	52.7	56.6	46.3	37.1	24.1	17.2
	2%	DB	2.4	1.1	6.5	18.3	32.5	51.5	61.8	57.0	47.0	31.1	21.7	12.5
		MCWB	1.8	0.7	5.9	17.6	31.5	45.5	51.2	49.6	41.9	30.8	21.3	11.9
	5%	DB	-6.1	-5.1	1.8	13.8	29.6	47.9	59.5	51.6	43.0	27.4	16.3	5.7
		MCWB	-6.8	-5.7	1.1	13.1	28.9	43.3	50.0	46.8	40.0	26.9	15.6	5.1
	10%	DB	-13.1	-11.0	-3.8	10.4	26.8	43.7	56.6	46.7	39.1	24.2	9.5	0.6
		MCWB	-13.6	-11.4	-4.4	9.6	26.1	40.3	48.2	43.1	37.6	23.7	8.8	-0.1
	0.4%	WB	17.7	8.6	12.2	24.0	34.6	49.2	54.6	56.8	46.9	37.1	24.3	18.5
		MCDB	18.1	8.9	12.7	24.3	35.4	53.7	62.3	64.1	51.7	37.4	24.6	18.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	2.1	6.1	17.9	31.5	46.1	52.3	50.3	42.7	31.1	21.4	12.6
		MCDB	4.1	2.6	6.5	18.3	32.2	51.2	60.2	55.3	46.4	31.2	21.8	13.5
	2%	WB	-2.7	-3.2	1.9	13.2	29.1	43.2	50.5	46.6	40.1	27.1	15.8	6.1
		MCDB	-2.3	-2.8	2.5	13.7	29.8	47.5	58.4	51.2	42.4	27.5	16.6	6.7
	5%	WB	-8.7	-8.2	-3.4	9.6	26.1	40.5	48.5	43.0	37.5	23.9	8.8	1.6
		MCDB	-8.1	-7.7	-2.8	10.4	26.7	43.4	55.7	45.9	39.5	24.3	9.5	2.1
	10%	WB	-8.1	-7.7	-2.8	10.4	26.7	43.4	55.7	45.9	39.5	24.3	9.5	2.1
		MCDB	-8.1	-7.7	-2.8	10.4	26.7	43.4	55.7	45.9	39.5	24.3	9.5	2.1
	5% DB	MDBR	8.1	8.6	9.0	8.9	6.8	6.4	8.7	7.4	5.5	7.1	8.2	7.9
		MCDBR	15.1	11.1	10.5	10.8	7.9	11.7	11.4	14.1	7.6	6.8	11.7	9.9
		MCWBR	14.3	11.0	10.3	10.8	8.0	8.5	6.8	9.7	5.8	6.9	11.7	9.7
		MCWBR	14.3	12.8	10.1	10.7	7.9	10.6	11.5	13.0	7.6	6.9	11.6	10.1
(35) Mean Daily Temperature Range	5% WB	MCWBR	14.0	12.6	10.0	10.7	8.0	8.2	7.7	9.4	6.2	7.1	11.7	9.9
		MCWBR	14.0	12.6	10.0	10.7	8.0	8.2	7.7	9.4	6.2	7.1	11.7	9.9
		MCWBR	14.0	12.6	10.0	10.7	8.0	8.2	7.7	9.4	6.2	7.1	11.7	9.9
		MCWBR	14.0	12.6	10.0	10.7	8.0	8.2	7.7	9.4	6.2	7.1	11.7	9.9
(40) Clear-Sky Solar Irradiance	taub	taub	0.153	0.198	0.241	0.305	0.316	0.301	0.335	0.309	0.269	0.222	0.147	N/A
		taud	1.862	2.075	2.172	2.011	2.069	2.235	2.361	2.474	2.443	2.120	1.900	N/A
	Ebn at Noon	Ebn at Noon	38	201	257	262	274	284	270	267	250	180	36	0
		Edh at Noon	2	14	24	39	42	36	31	25	19	13	2	0
(44) All-Sky Solar Radiation	RadAvg	RadAvg	10	157	676	1429	2014	2068	1612	1087	541	186	28	0
		RadStd	1	8	21	43	90	110	151	86	66	25	2	0

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page